



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : EE G508

Course Title : Urban Water Management

Instructor-in-charge : Prof. Rallapalli Srinivas

[Team of Instructors] : Prof. Rallapalli Srinivas

Timings : M W (4th: 11 am to 12 pm), T 10 (5 pm to 6 pm), 1233

1. **Course description:** Urban Water Management is an advanced course focusing on the planning, design, operation, and assessment of urban water systems under increasing pressures of population growth, urbanization, and climate change. It explores the complete urban water cycle, including supply, sanitation, drainage, and environmental flows, with strong emphasis on sustainable and resilient urban development. The course addresses technical, social, economic, and regulatory challenges through a multidisciplinary lens. Students are exposed to contemporary topics such as stormwater management, water-sensitive urban design, and water quality monitoring. Practical software-based training modules on stormwater modelling, flood forecasting, and hydroinformatics are integrated to bridge theory and real-world applications. Students will also gain insights into advanced asset management systems, sustainability indicators, GIS tools, and the effect of urbanization on hydrological behavior. This industry-oriented course prepares students for professional roles in government, consulting, and research sectors concerned with smart, equitable, and sustainable urban water solutions.
2. **Scope and Objective:** The course aims to equip students with comprehensive knowledge and practical skills in managing urban water systems effectively and sustainably. The scope spans water supply, sanitation, stormwater, groundwater, and urban drainage management. It incorporates key concepts in water-sensitive urban design, reuse and recycling of wastewater, climate-resilient infrastructure planning, and integrated urban water modelling. Emphasis is placed on understanding emerging urban water challenges, regulatory frameworks, and digital tools for

decision support. Through hands-on sessions, case studies, and software exposure, the course objectives are: (1) To understand the integrated urban water cycle and its components; (2) To develop competence in water quality and infrastructure assessment; (3) To analyze risks and sustainability indicators in urban water systems; (4) To apply hydroinformatics tools for modeling and performance evaluation; and (5) To foster decision-making skills relevant to urban water management, policies, and climate adaptation strategies.

3. Course learning outcomes:

- Understand and assess the urban water cycle and system interactions under climate change and urbanization.
- Demonstrate practical knowledge in water quality, stormwater, and sanitation system design and management.
- Use hydroinformatics and modeling tools for flood forecasting and system performance evaluation.
- Integrate water-sensitive urban design principles and sustainability indicators in planning.

4. Text books and references:

TB1. Water Supply Engineering, S. K. Garg (18th ed.), Khanna Publishers.

TB2. Water Supply and Sanitary Engineering, G. S. Birdie & J. S. Birdie (8th ed.), Dhanpat Rai Publishing Company, New Delhi.

R1: STORMWATER DRAINAGE MANUAL Planning, Design and Management, Drainage services department, Government of the Hong Kong Special Administrative Region.

R2: Peavy, H. S., Rowe, D. R., & Tchobanoglous, G. (1985). Environmental Engineering. McGraw Hill Education (India Edition, 2013).

R3: Garg, S. K. (2010). Sewage Disposal and Air Pollution Engineering: Environmental Engineering (Vol. 2). Khanna Publishers, Delhi.

5. **[Method of conduct of the course]:** The course will be conducted through a combination of interactive lectures, software-based tutorials, case study discussions, and hands-on modeling sessions. Emphasis will be placed on blending theoretical foundations with practical applications to address real-world urban water challenges. Students will engage in:

- Lectures using multimedia tools to explain concepts related to the urban water cycle, supply, drainage, and quality management
- Software training modules on stormwater modeling, urban flood forecasting, and hydroinformatics using tools such as SWMM, EPANET, HEC-RAS, and QGIS.



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- Group assignments involving real-world problem-solving, such as urban flood mitigation planning or rainwater harvesting assessment.
- Guest lectures or webinars by industry/academia/government experts and municipal engineers to expose students to professional practices.
- If possible, field visits (where feasible) to treatment plants, stormwater systems, or urban recharge zones for experiential learning.
- Project-based evaluations including presentations, technical reports, and modeling outputs aligned with industry standards.

6. Course Plan: weekly schedule for 14 weeks with [3] lectures per week

Module	Week(s)	Lecture	Reference	Learning Outcomes
Module 1: Introduction to Urban Water Systems	Week 1-2	[L1-L6]	TB1, TB2	- Understand the structure and significance of the urban water cycle. - Identify social, environmental, and economic challenges in urban water management.
Module 2: Urban Water Supply and Sanitation	Week 3-5	[L7-15]	TB1, TB2	- Explain water sourcing, distribution systems, and recycling strategies. - Evaluate water quality parameters and understand testing and treatment technologies. - Understand fundamentals of sanitation, sewerage, and urban drainage design.
Module 3: Stormwater and Groundwater Management	Week 6-8	[L16-24]	TB1, TB2	- Analyze stormwater drainage systems and sustainable urban drainage systems (SUDS). - Design recharge and rainwater harvesting systems in urban areas. - Examine groundwater challenges and risk management in urban contexts.
Module 4: Sustainable Design and Decision Tools	Week 9-10	[L25-L30]	TB1, TB2	- Apply water-sensitive urban design principles and sustainability indicators. - Understand fundamentals of asset management and GIS applications in water systems.

Module 5: Software Applications and Advanced Topics	Week 11-14	[L31-L42]	USEPA, journal papers	- Use modeling tools for urban stormwater and flood management (e.g., SWMM, HEC-RAS). - Apply hydroinformatics tools for advanced systems performance assessment. - Evaluate the effects of urbanization and climate change on urban water systems. - Synthesize learning through case studies and project-based assessments.
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Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
[Class discussions/Assignments]	[10%]	[In class]	[In class]	[Open book]
[Handson software usage in class]	[10%]	[In class]	[In class]	[Open book/Open-laptop]
[Project]	[20%]	[3 months]	[In class]	Presentations and reports
[Mid Semester]/Software demonstration and problem solving in class	[25%]	90 min	5/10 AN2	[Open/close notes]
Comprehensive	[35%]	180 min	3/12 AN	[Open/close notes]

Criterion for NC:

7. Each student is expected to attend all classes and participate in all evaluation components.
8. Chamber consultation hours: Friday 12 noon -1 pm, 4 pm to 5 pm
9. Makeup Policy: Only in extreme medical emergency

Prof. Rallapalli Srinivas

Instructor-in-charge



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